



D3.2 UPPER-LEVEL REGULATORY SECTORAL ONTOLOGIES & VOCABULARIES – ITERATION 1

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Iteration 1

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Abstract	This report examines the critical role of semantic interoperability in enabling an effective Digital Product Passport (DPP) system, analyzing data models, ontologies, and classification schemes across electronics, textiles, tyres, and construction. While semantic anchors provide a foundation, reliance on a single standard can create conflicts, necessitating a flexible, multi-ontology approach for seamless data integration. Ongoing collaboration with standardization bodies, industry stakeholders, and regulators is essential to maintain adaptability across diverse value chains. This iteration maps existing ontologies and identifies gaps. The next phase, which will refine sector-specific ontologies and ensure harmonized interoperability between Lighthouse pilots.
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43	Physikalisch-Technische Bundesanstalt	PTB	Germany
44	Digital Data Chain Consortium GbR	DDCC	Germany
45	ZVEI e. V.	ZVEI e.V.	Germany
46	Association of Service and Computer Dealers International	ASCDI	US
47	Open Blockchain for Asset Disposition Alliance (OBADA)	OBADA	US
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LIST OF ABBREVIATIONS AND ACRONYMS

List of Abbreviations and Acronyms	
API	Application Programming Interface
CEN	European Committee for Standardisation
CENELEC	European Committee for Electrotechnical Standardisation
DPP	Digital Product Passport
EC	European Commission
EPCIS	Electronic Product Code Information Services
ERP	Enterprise Resource Planning
ESPR	Ecodesign for Sustainable Products Regulation
GTIN	Global Trade Identification Number
IEC	International Electrotechnical Commission
ISO	International Organization for Standardization
PIM	Product Information Management
PLM	Product Lifecycle Management
POS	Point of Sale
REO	Responsible Economic Operator (legally responsible for issuing DPPs)
RFID	Radio-Frequency Identification (type of tag)
SGTIN	Serialized Global Trade Identification Number
UID	Unique Identifier
UPI	Unique Product Identifier
VC	Verifiable Credential

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1 INTRODUCTION

This report focuses on existing data models, ontologies and vocabularies which may be required for the implementation of Lighthouse pilots across electronics, textiles, tyres and construction value chains. It builds upon a previous state of art presented in the CIRPASS report “DPP Prototypes”¹. To ensure this document remains valuable and up-to-date, experts in these topics are encouraged to contribute suggestions for improvements

Due to their critical role in electronic data exchange, data models and semantics often become battlegrounds for organizations seeking to impose their own systems, potentially hindering progress and collaboration. Achieving semantic interoperability is essential to overcoming these challenges, ensuring a more efficient and harmonized Digital Product Passport (DPP) system.

2 THE ROLE OF SEMANTIC INTEROPERABILITY IN THE DPP

The essential role of semantic interoperability in the context of the DPP, can be derived from highlighted text in the Wikipedia definition below.

“Semantic interoperability is the ability of computer systems to exchange data with unambiguous, shared meaning. Semantic interoperability is a requirement to enable machine computable logic, inferencing, knowledge discovery, and **data federation between information systems**. Semantic interoperability is therefore concerned not just with the packaging of data (syntax), but the **simultaneous transmission of the meaning with the data** (semantics). This is accomplished by adding data about the data (metadata), linking each data element to a **controlled, shared vocabulary**. The meaning of the data is transmitted with the data itself, in one **self-describing "information package" that is independent of any information system**. It is this shared vocabulary, and its associated links to an ontology, which provides the foundation and capability of **machine interpretation**, inference, and logic.”²

According to the European Commission, “semantic interoperability ensures that the precise format and meaning of exchanged data and information is preserved and understood throughout exchanges between parties”.³ In other words, the meaning of the exchanged data remains intact, and understood by all parties.

¹ Bernier, C., Danash, F. (2024) Digital Product Passport (DPP) Prototypes, CIRPASS Consortium, <https://doi.org/10.5281/zenodo.11393742>

² https://en.wikipedia.org/wiki/Semantic_interoperability (accessed January 8, 2025)

³ <https://interoperable-europe.ec.europa.eu/collection/nifo-national-interoperability-framework-observatory/solution/eif-toolbox/interoperability-layer-5-semantic-interoperability>

3 SEMANTIC INTEROPERABILITY BUILDING BLOCKS

According to the European Commission's Semantic Interoperability Courses⁴, semantic interoperability is achieved when social agreements are reached on vocabularies (common specifications for naming things) and on structural meta data (data models and reference data). They define several key terminologies:

Metadata is data that defines and describes other data (ISO/IEC 11179-1). Structural metadata gives meaning to data and indicates how it is structured. It typically consists of **Reference data**, a small, discrete set of values that are usually used to impose consistent classification and a data model.

A **data model** is a collection of entities, their properties and the relationships among them, which aims at representing a domain, a concept or a real-world thing. A data model contains:

- **Classes:** the distinct types of things that exist in our data, typically organized in a class hierarchy.
- **Relationships or Object properties:** properties that connect two classes, typically organized in a property hierarchy.
- **Data properties (or attributes):** properties that describe an individual class and are thus relations between a class and a literal.

Data models are typically designed for a specific application. There is a vast choice of tools that can be used to design a data model, from the simplest (a spreadsheet) to the most complex (an industrial digital twin).

Ontologies are **formal data models** designed for greater generality and expressivity. Similarly to data models, ontologies define the types of things that exist in a given domain and the properties that can be used to describe them. When classes and properties are combined, the ontology can be viewed in a graph format. Because they are generalized data models, ontologies only model general types of things that share certain properties but don't include information about specific instances. Ontology-based approaches guarantee the structuring of data and the effective use of derived knowledge, which includes tasks like reasoning over data.

In contrast from the data models which can be implemented with a vast choice of tools, **ontologies are data models with a standardized technical representation**.

3.1 USES OF SEMANTIC INTEROPERABILITY BY THE EUROPEAN COMMISSION

Since 2014, semantic interoperability has been increasingly promoted by the European Commission to help public administrations in Member States to perform seamless and meaningful, cross-border and cross-domain data exchanges.

The example below demonstrates how concepts from an existing ontology (here schema.org, illustrated in dark grey), can be reused and extended for new purposes. Figure 1 illustrates an example of how legislation is described. The new concepts are in blue, defined as 'SubClassOf' existing concepts, and with additional properties. The legislation described in Figure 1 has been achieved by using a range of generic attributes

⁴ Module 1 – Introduction and overview of existing initiatives, ISA Programme, Action 1.1 https://joinup.ec.europa.eu/sites/default/files/document/2014-06/Semantic%20interoperability%20courses%20-%20Training%20Module%201%20-%20Introductory%20overview_v0.19.pdf

available in the CreativeWork class, and by using specific attributes defined in the European Legislation Identifier (ELI) ontology.

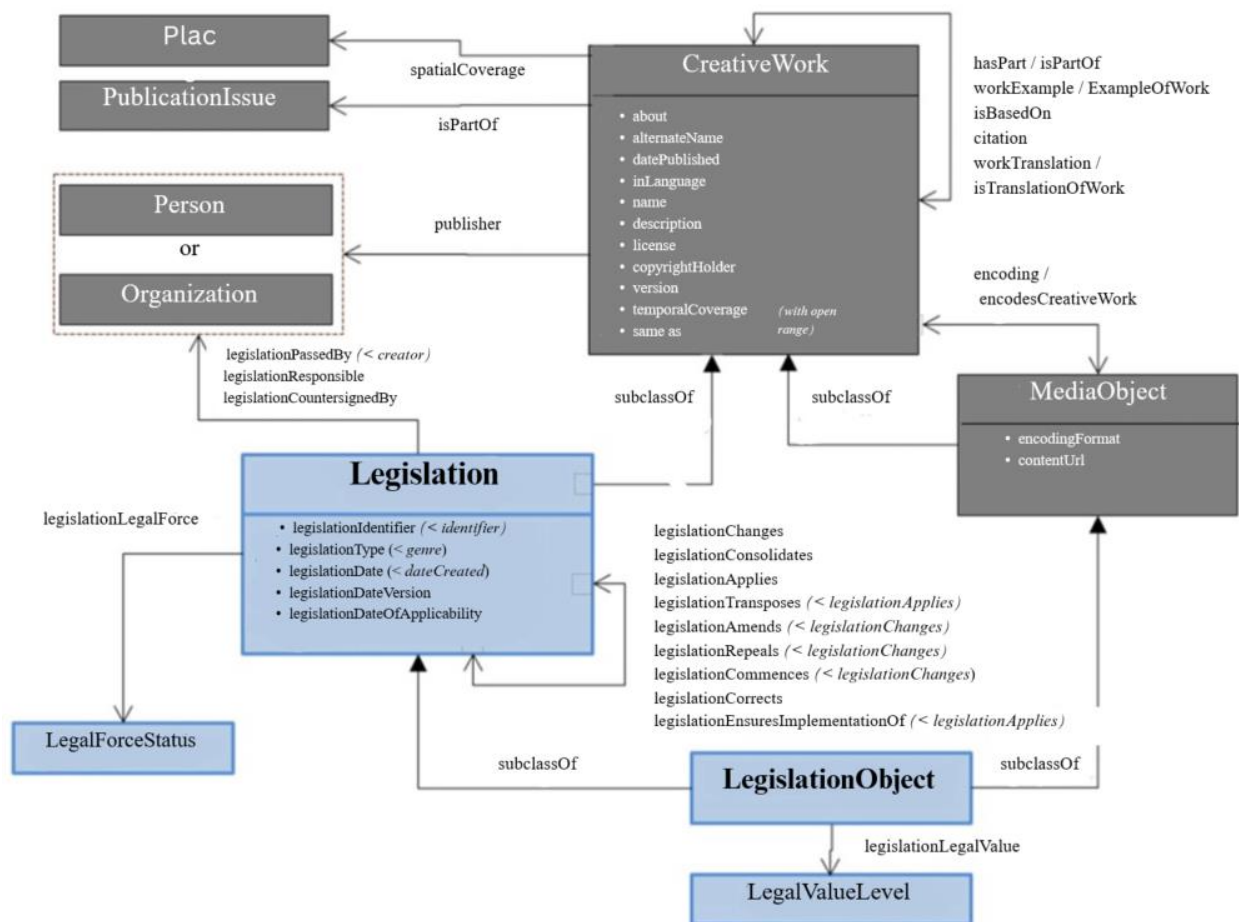


Figure 1. Example of an ontology, here the European Legislation Identifier (ELI) ontology (blue) which extends the schema.org ontology (dark grey). Source https://eur-lex.europa.eu/eli-register/legis_schema_org.html#eli-ontology-guides

3.2 THE PROBLEM WITH SEMANTICS

Organizations across various levels—such as economic operators, industry associations, and international bodies—invest significant effort in developing semantic frameworks tailored to the needs of IT systems (as illustrated below). Depending on the level of consensus required to agree on the semantics, these efforts can span over a number years. Once implemented, information systems are then typically bound to the chosen semantics. As a result of the effort required to manage any alterations to these information systems, organisations tend to:

- resist any changes in semantics and,
- attempt to impose their own semantics on others.

As a result, the capability to reuse existing semantics is crucial.

The table below illustrates the increasing effort required to develop semantic frameworks for IT systems, from creating local data models to establishing industry-wide standards and ontologies. While tasks like

standardizing information points demand significant resources, converting these frameworks into machine-readable, interoperable formats requires comparatively less effort. This highlights the growing complexity involved in achieving digital interoperability.

Activity	Effort (time)
Developing a data model (local to one information system)	+ +
Developing a data model for an industrial ecosystem	+ + +
Developing dictionaries, classification systems	+ + + +
Developing a domain ontology (concepts, relations)	+ + + +
Developing standards for information points (product carbon footprint, durability, ...)	+ + + +
Converting the above into machine readable, semantically interoperable formats	+

Table 1: Levels of Effort Required for Developing Semantic Frameworks in IT Systems

3.3 THE ROLE OF SEMANTIC ANCHORS

The resistance to change highlighted in the results is also evident in **translation complexity** (Figure 2, left). Each organization employs unique data semantics within its IT systems, requiring customized electronic data exchanges. This complexity often hinders the transition from manual to electronic processes. **Semantic anchors**, such as ontologies or standardized data models, help reduce mapping challenges.

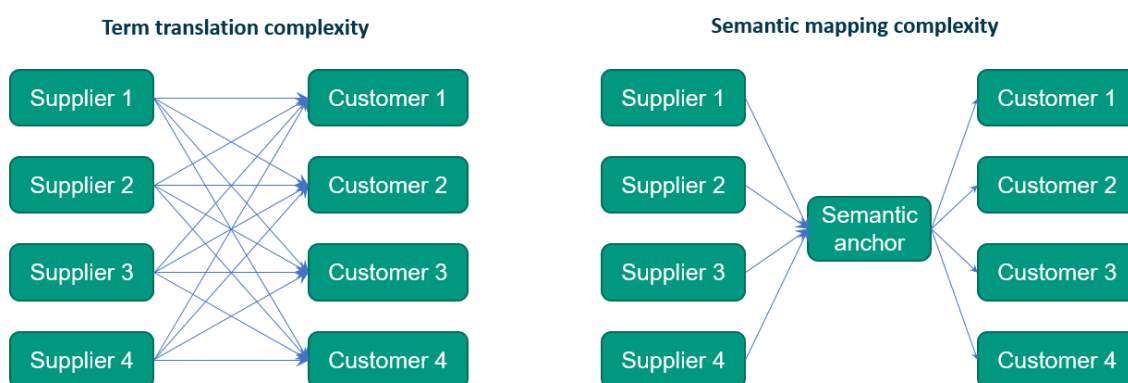


Figure 2. The role of semantic mapping to limit term translation complexity

However, using data standards does not eliminate conflicts, as differing standards create a **standard-vs-standard** challenge rather than an **organization-vs-organization** issue. Resolving this requires **semantically interoperable anchors** (Figure 3, right), which enable seamless integration across standards. **Semantic technologies** provide a viable solution to overcome these conflicts.



Figure 3. Avoiding the unification trap through the use of semantically interoperable anchors

3.4 AN ONTOLOGY WORLDVIEW FOR THE DPP SYSTEM

The cross-sectoral DPP system ontology illustrated in Figure 4 should enable the anchoring of terms to universally applicable concepts across all product groups. By providing a common semantic foundation, this ontology ensures that data is structured, interpreted, and exchanged consistently across industries. However, rigid reliance on a single semantic anchor can introduce conflicts rather than resolving them, as discussed earlier. To avoid this ‘single semantic anchor trap,’ the ontology must be designed for semantic interoperability, allowing it to integrate with and adapt to multiple existing ontologies.

Finally, semantic interoperability plays an essential role in facilitating the use and re-use of existing ontologies, and domain vocabularies from various sectors. By leveraging internationally recognized standards, it can ultimately have the effect of reducing complexity and improving data exchange efficiency within the DPP ecosystem.

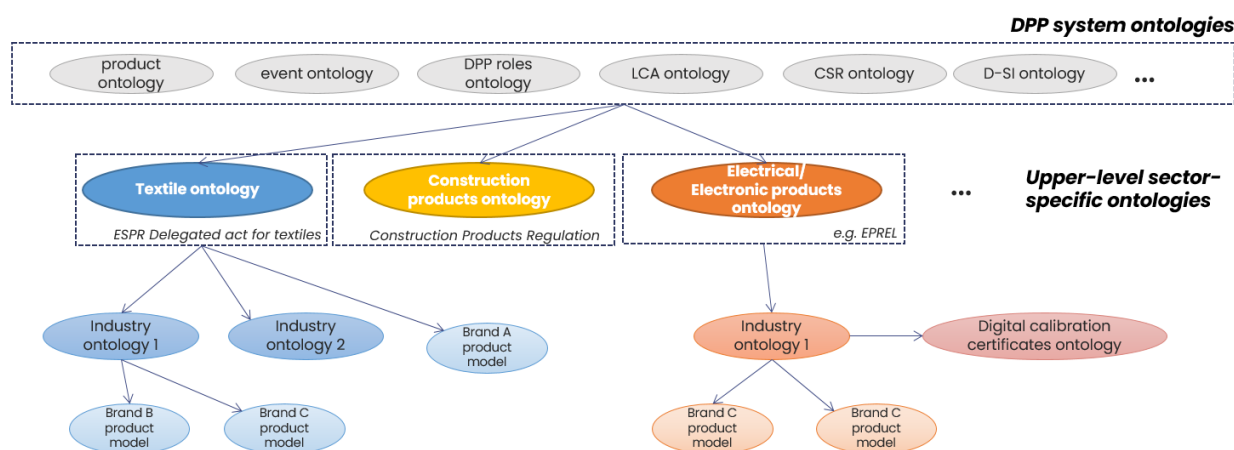


Figure 4. Enabling cross-sectoral semantic interoperability in DPP system

4 EXISTING DATA MODELS & ONTOLOGIES: TEXTILE PRODUCTS

This initial iteration expands on the Cirpass-1 deliverable “5.1 DPP Prototypes” by incorporating ongoing initiatives, and by introducing a preliminary structure. The listing is compiled based on contributions from task members, without imposing any “minimum requirements” (such as open documentation, format, relevance, or usage) for inclusion at this stage. This section remains open for feedback from project participants; and external experts will be consulted for additional review before the next iteration.

4.1 ONTOLOGIES

Below is a preliminary list of data models and ontologies relevant to the textiles sector:

TRICK/eBIZ

The EU-funded **TRICK** project, which is defining a traceability platform to enable verifiable claims related to textiles circularity and other sectors, recently published a review of data models and ontologies related to the textiles sector [TRICK 2022, Chapter 4.2.1]. From this survey, they have published an ontology, based on the eBIZ specification framework, adapted to their needs and which may have considerable synergies with the DPP [TRICK 2022b]. Based on the results of the TRICK project, a CEN Workshop Agreement was launched in December 2024 to establish guidelines aimed at enabling and optimizing data collection along textile supply chains, with the objective of improving traceability, transparency, and sustainability claims, in line with European regulations.

United Nations Transparency Protocol (UNTP)

The United Nations Transparency Protocol (UNTP) aims to support governments and industry with practical measures to counter greenwashing by implementing supply chain traceability and transparency at the scale needed to achieve meaningful impacts on global sustainability outcomes.

The UNTP provides a solution to the transparency challenges facing the world's supply chains. By implementing a simple protocol that can be supported by existing business systems, stakeholders will realise immediate benefits and will become visible contributors to the sustainability of global supply chains.

The UNTP development follows the same standard governance rules as any UN/CEFACT project: free to use, open source licensed, maintained via an open process, version controlled, and lifecycle managed.

UNECE

To date, the most comprehensive overview of data standards and classifications used within the textile industry is the Reference Guide on Code Lists and Identifiers in the Textile and Leather Value Chains published by UNECE in October 2022⁵. Its purpose “is to identify and describe code lists and identification schemes supporting the business processes and transactions for traceability and transparency in the textile and leather value chains.” The UNECE Reference Guide shows that there is considerable coverage and overlap, but, given the publication date and the speed of change in the industry, standards focusing e.g. on circularity were missing. Furthermore, several sustainability-focused standard and certification systems which have means of identifying and storing data of certified products, facilities or processes, are not publicly available and are therefore not included.

The **UNECE-UN/CEFACT**, through its recent “Product Circularity Data Use Case for textiles and leather – Garments and footwear products” project, has defined a Conceptual Product Circularity Data Model

⁵ https://unece.org/sites/default/files/2023-10/Guidelines-ECE_TRADE_C_CEFACT_2022_INF1E.pdf

focusing on tracking and tracing for circularity. [UNECE 2024] The focus of the information model is on data supporting circular business models for a circular economy. It is meant to be seamlessly integrated into the scope of the DPP, as shown in **Erreur ! Source du renvoi introuvable.** It is also meant to be used over the full value chain, from cradle to grave, with the exchange of product circularity data over the pre-product-consumption phase of the supply chain to the post product-consumption phase.

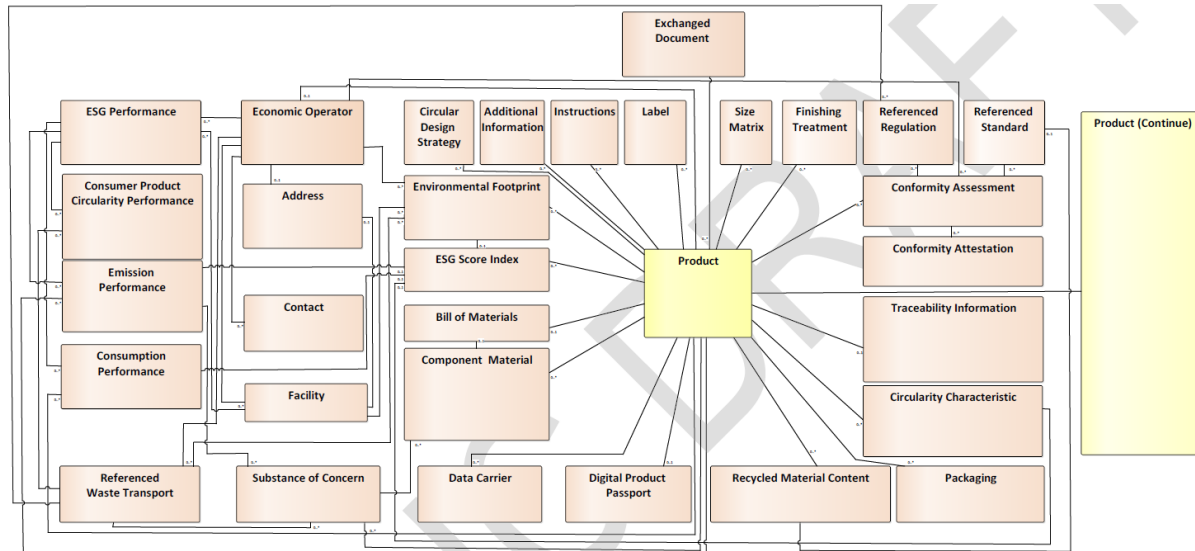


Figure 5. Conceptual product circularity data model reproduced from [UNECE 2024]

This model includes the definition of many circular economy actor roles, as shown in the table below, but only 2 business process roles for the actual data exchange, the Requestor and the Responder:

Business Process Role	Product Circularity Data Requestor Product Circularity Data Responder
Supporting Role	Farmer, Breeder, Spinner, Weaver, Designer, Manufacturer, Subcontractor, Brand Owner, Retailer, Trader, Wholesaler, Waste Exporter, Waste Importer, Governmental Authority, Recycler, Waste Collector, Waste Aggregator, Finishing provider, Tanner, Raw fibre treatments provider, Slaughterhouse, Warehouse, Transporter, Third Party Reseller, Product Identity Platform, Repair Provider, Retailer, Online Market Place, Agent, Consumer, Waste Sorter, Refurbishment Centre / Refurbisher, Collection Centre, Waste Disposal Provider, Third Party Manufacturer, Waste Sorter, Waste Incineration Facility, Landfill Site, Industrial Composting Facility, Waste Pre-Processor, Other Supplier, Tier-1-n Supplier/Manufacturer, Provider of IDs, Customer-Consumer

Figure 6. Circular economy actor roles defined in [UNECE 2024]

Most of the standards that are publicly available have either machine-readable code list (short code with term) or are organized in formal data structures (e.g., XML, HTML, spreadsheet). Many have both. However, many of the largest and most widely used standards are kept behind a paywall, and so are not readily accessible to all users. Finally, often the term “classification system” is used instead of “data model” or “ontology” which creates uncertainty as to whether the approach is actually used to describe the properties of products or to simply to classify them. However, even product classification schemes provide general descriptions of eligible items for each category, even if this is in an unstructured form. For this reason, a number of them are included below, an unfortunately far from exhaustive list which focuses on Western Europe.

ISO/IEC 19987

ISO/IEC 19987: The goal of EPCIS is to enable disparate applications to create and share visibility event data, both within and across enterprises. Ultimately, this sharing is aimed at enabling users to gain a shared view of physical or digital objects within a relevant business context.

ISO/IEC 19988:

ISO/IEC 19988: The goal of this standard is to specify various vocabulary elements and their values for use in conjunction with the EPCIS standard [EPCIS2.0], which defines mechanisms to exchange information both within and across organisation boundaries. EPCIS and the CBV are developed, maintained and published by GS1; EPCIS and the CBV are also published within ISO's PAS process as ISO/IEC 19987 and ISO/IEC 19988, respectively. The vocabulary identifiers and definitions in this standard will ensure that all parties who exchange EPCIS data using the CBV will have a common understanding of the semantic meaning of that data.

circularity.ID

The circularity.ID Open Data Standard and ontology for textiles and fashion, Version 4.0 (soon V5.0) was developed by circular.fashion for circularity in textiles and fashion in collaboration with a wide group of industry stakeholders. It provides a data protocol and ontology to make product data available for circularity checks and for textile sorting, reuse, and recycling.

Circular Product Data Protocol

The Circular Product Data Protocol (CPDP) was developed by a large consortium of industry stakeholders under the leadership of EON and proposes a circularity-focused data model for product and material-level data. The CPDP calls upon and leverages existing standards from GS1, Sustainable Apparel Coalition, Open Supply Hub, UNECE, etc.

4.2 WEB VOCABULARIES

Integrated with Schema.org, GS1 is maintaining vocabularies for [Clothing](#) and [Footwear](#).

4.3 CLASSIFICATION SCHEMES

This section outlines key classification schemes used across the textile, footwear, and fashion industries, highlighting their role in standardizing product categorization, supporting supply chain management, and facilitating industry-wide interoperability.

- The **EAS** (European Article System) is used for shoes by most of the shoe industry and retailers. It is maintained by associations and paid by membership fees.
- The **FEDAS** classification system is widely used for sporting goods and is accepted by BTE (Bundesverband des Deutschen Textil-, Schuh- und Lederwareneinzelhandels – Federal Association of the German Textile, Shoe and Leather Goods Retailers). A revision is planned in 2024.
- The **BTE** classification system is used by many retailers for fashion retail mainly in German-speaking countries.
- **Refashion**, the French textile industry's eco-organisation for clothing, household linen and footwear, has a classification of product categories.⁶

⁶ Refashion classification of product categories, URL: <https://refashion.fr/pro/sites/default/files/fichiers/Product Classification File - textile-EN V2.xlsx>

- The **ICECAT** classification system is provided by the product data platform provider of the same name for different articles including fashion in the Netherlands.
- **Textile Exchange** has a material classification system (ASR-213), which aims to provide standardized codes for raw materials, processes, product categories and product details for the textile supply chain.⁷ In the fall of 2024, Textile Exchange and Apparel Alliance completed work on a Supply Chain Taxonomy for the textile, apparel and fashion industry⁸. Similar to ASR-213 (materials, processes, and products), this document aims to establish a uniform classification system of the textile supply chain.
 - Note 1: Raw Materials used for label claims in EU must adhere to the Textile Labeling Regulation which is currently being revised. A draft is expected in June 2025, and a final version sometime in 2026. TE's classification is based on that and will also need to be revised when the update is released.
 - Note 2: There is currently discussion on the use of 4-digit Combined Nomenclature (CN) codes for the purpose of reporting unsold goods as required by ESPR.
- **Cascale/Higg classification**⁹: The Higg Index is a suite of five tools that assess and measure the social and environmental performance of the value chain and the environmental impacts of products. They developed these tools to help organizations make systematic change by identifying, understanding, and measuring areas of improvement.
- **Open Supply Hub Classifications**: The OS ID is a unique Reference Number which is allocated to a specific manufacturing site and held within the Open Supply Hub database (OS Hub) including a classification for the textile facilities and the related activities.
- **GOTS** (Global Organic Textile Standard) and Textile Exchange jointly developed a classification on materials, processes, and products for the harmonized application of their policies for scope and transaction certificates.¹⁰
- The **GINETEX** care label standard is a material classification scheme.
- The Technical Secretariat of the **PEFCR** for Apparel and Footwear, including associations and producers representing jointly more than 51% of the EU apparel & footwear market, introduced performance requirements, Environmental footprint (LCA) modelling, data and reporting requirements for PEF studies on 13 sub-categories covering most types of apparel and footwear products. Also, the underlying reference data of the European Commission's Environmental footprint. The work is anticipated to be finalised in Q1/Q2 2025.
- The **Global Textile Scheme** (GTS) classification system and meta-standard has been available since summer 2023. It includes a data model with well-defined semantics and applies to all sectors of the textiles value chain. Its main use is the translation of existing product data of a data sender into a machine readable format which can be decoded by the data receiver into their own natural language and mapped to their own data formats. GTS integrates several classification systems such

⁷ Textile Exchange, Materials, Processes, & Products Classification — ASR-213-V1.1-2021.05.01, URL <https://textileexchange.org/app/uploads/2020/10/ASR-213-V1.1-Materials-Processes-and-Products-Classification.pdf>

⁸ <https://textileexchange.org/app/uploads/2024/12/Supply-Chain-Taxonomy.pdf>

⁹ <https://howtohigg.org/higg-msi/guidance-and-definitions/content-guidance-overview/>

¹⁰ Global Standard GmbH & Textile Exchange: Materials, processes & products classification, version 1.0, May 2021, URL: https://global-standard.org/images/resource-library/documents/certificate-policies-and-templates/Materials_Processes_and_Products_Classification_v_1.0.pdf

as EAS, FEDAS, BTE, and DTB. The raw material classification of GTS is aligned to the raw material classes of Textile Exchange, GINETEX, Afirm Group and the latest product class groupings by JRC “duration publishment” from November 2024. GTS covers components, demand data (between brands and suppliers) and offers the management of a broad range of product-related certificates.

- The **CISUTAC** project has recently published a decision support tool¹¹ providing recommendations on necessary datapoints for textile sorting. The tool uses various post-consumer textile product attributes such as condition, construction, chemical content, etc. to comprehensively evaluate the optimal route to either reuse, repair or recycling for the post-consumer product.

¹¹ <https://www.cisutac.eu/solution-post-consumer-textile-waste> (accessed April 9, 2024)

5 EXISTING DATA MODELS & ONTOLOGIES: ELECTRONICS PRODUCTS

The section provides an overview of relevant classification standards including data dictionaries as well as developed ontologies for electrical and electronic products. Broader initiatives are listed first in the section, and specific ontologies such as used for particular products are defined in Table 2 below.

5.1 ONTOLOGIES

RePlanIT

RePlanIT is an ontology for the sharing of ICT product data between manufacturers, sustainability experts and technology providers for the circular economy in view of future DPPs for ICT devices [Kurteva 2024]. The RePlanIT ontology for ICT DPPs captures knowledge on several levels - ICT device, hardware components, materials and the circular economy itself. RePlanIT's specification is based on a literature survey, interviews and inputs from domain experts from both industry and academia. The use of the ontology has been successfully validated in a real-world scenario.

eReuse.org

eReuse.org, a program originally funded by the European Union with the goal of improving the circularity of electronic devices, developed an ontology developed by UPC in collaboration with social ICT refurbishers and recyclers in Catalonia [Talavera 2016]. This ontology is used in a software architecture that supports the management of second hand devices (such as desktops, laptops, and smartphones) and creates a global traceability standard for ICT devices.

New initiatives for electrical and electronic products:

ITU-T/ETSI L. L.D4PI

The ITU-T/ETSI L. L.D4PI¹² work group is currently elaborating a draft recommendation “An information model for digital product information on sustainability and circularity”. The proposed Recommendation will provide a collection of information items organised to represent circularity, environmental sustainability and health information about ICT products and inform any actor during the product lifespan: design, manufacturing, usage phases including changes over the lifespan, until final recycling as e-waste. This work will determine the information items, to be represented in digital format, about ICT product information, either as individual products or grouped by model, batch, etc., common to all ICT products or specific to certain product categories. The objective is that this standard is technically aligned with corresponding work at ETSI. This standard will undoubtedly be linked to the ITU-T L.1023 standard on “Assessment method for circular scoring” which proposes criteria for the assessment of circularity and definitions of margins of improvement levels of ICT goods.

CE-RISE

The CE-RISE (Circular Economy Resource Information System)¹³ Horizon Europe project aims to create an information system and integrate digital product passports to share detailed information on electronic

¹² https://www.itu.int/ITU-T/workprog/wp_item.aspx?isn=18559

¹³ <https://ce-rise.eu/>

products. The purpose is to provide stakeholders, including consumers, with a better understanding of the green credentials of electronic products and how to preserve critical raw materials through the reuse, repair and recycling of these items. To this end, the project is actively developing a vocabulary in order to create a common language to specify the circular properties and circular strategies for products.

PEFCR

PEFCR for “IT equipment” (in fact: data centre-scale hard-disk systems), and PEFCR for PV, and the underlying reference data of the European Commission’s Environmental footprint and related information/approach for functional units, LCA/EF modelling, and performance related measurement standards to be used.

ONTOLOGIES FOR SPECIFIC ELECTRONIC PRODUCTS

Ontology	Access	Scope	Applications	Limitations	Reference
OneM2M	Private	ICT; laptops	Used for modelling laptop information for online sales	Focus on laptops in e-commerce only. Specific URIs, namespaces and object properties not included.	V. Dhingra and K.K. Bhatia, Development of ontology in laptop domain for knowledge representation, <i>Procedia Computer Science</i> 46 (2015), 249–256.
DogOnt	Public	Smart devices	Used to assist in auto-completion mechanisms based on reasoning	Focus on appliances that are IoT connected. Properties missing domain and range, unconnected concepts, inconsistent reuse and naming of concepts	D. Bonino and F. Corno, Dogont-ontology modelling for intelligent domotic environments, in: <i>International Semantic Web Conference</i> , Springer, 2008, pp. 790–803
Ayundhita et al.	Private	Laptops	Improving laptop retail recommendations	Focus only on laptops and information that can help to improve online retail. Not available online and not documented	Ayundhita, Z. Baizal and Y. Sibaroni, Ontology-based conversational recommender system for recommending laptop, in: <i>Journal of Physics: Conference Series</i> , Vol. 1192, IOP Publishing, (2019), p. 012020
Corcho et al.	Public	Hardware	Used as a basis for knowledge graphs that assists questions answered by a chatbot	High-level ICT representations. A complex ontology network that although built in a modular manner has many interdependencies.	O. Corcho, R. Alcazar, D.C.-F.J. Toledo, J. Arenas, M. Wang, H. Peng, N. Burrett, J. Mora and P. Zhang, Ontology for the representation of hardware items related to DevOps infrastructure (2021). Available at http://w3id.org/devops-infra/hardware .
Yowe and Astawa	Private	Laptops	Laptops and their components	Limited access to the ontology. Unknown evaluation	A.C.G. Yowe and I.G.S. Astawa, OntologyBased Approach For Laptop Semantic Knowledge Representation, <i>Journal Elektronik Ilmu Komputer Udayana</i> (2021)

Table 2. Examples of additional, specific ontologies for electronic products.

5.2 DATA DICTIONARIES

- **IEC Common Data Dictionary** (IEC CDD) is an international series of Standards for semantic properties in the form of structured descriptions or online databases (IEC 61360-4 DB, IEC 61987 series, IEC 62720, IEC 62683, IEC 63213) of concepts for all industrial/technical domains (electrotechnical and non-electrotechnical, e.g. industry, building, energy, healthcare, ...) based on the information model of IEC 61360-2. This dictionary may be used to define ontologies for use in the field of electrotechnology, electronics and related domains. This dictionary is currently being revised to provide additional properties needed for the evaluation and improvement of the environmental impact of products throughout the supply chain (IEC TS 63058:2021).
- **APPLIA PI-standard (PI)** is a standard to structure electrical product data across 118 product groups as a data dictionary with data field mapping. It has been developed in the sector to provide uniform information from manufacturers to retailers about their products, such that retailers can receive similar information in the same structure for products they sell regardless of the brand. Data in the PI-Standard includes marketing information, product feature, technical information and European Energy Labelling Information.

5.3 CLASSIFICATION SCHEMES

- **ETIM¹⁴** is an international classification standard for technical products which also includes electronics finished goods. The ETIM product classification model provides the structure for standardized (technical) product data exchange between parties. The standard facilitates the listing and publishing of complete product data with the aim of making it easier to identify the technical products required by the user. ETIM has a flat organizational structure and the main focus is the definition of product classes and their features which are structured according to their importance as well as key aspects. A feature is clearly described by description, feature type, unit and/or value.
- **ECLASS¹⁵** is a standard for the hierarchical classification and grouping of products and services designed to support their unambiguous description and the digital exchange of product master data. ECLASS develops different standards from industries such as electrical engineering, automation, process control engineering, food, automotive and office supplies. The ECLASS Standard currently offers about 48,000 product classes and more than 23,000 unique properties that can be collectively categorized with only four levels of classification: Segment, Main Group, Group and Subgroup. ECLASS will soon publish an ECLASS as RDF specification, which will be the first official ECLASS as RDF specification including all ECLASS Structural Elements. This allows the usage of ECLASS in, e.g., W3C Web of Things (WoT) Thing Description (TD) and AAS as RDF.
- **NACE, HS, CN and other statistical classifications** on any kind of products and production processes/economic activities, used e.g. by customs authorities and for company reporting obligations.

¹⁴ <https://www.etim-international.com/classification/model-information/>

¹⁵ <https://eclass.eu>

6 EXISTING DATA MODELS & ONTOLOGIES: CONSTRUCTION PRODUCTS

6.1 ISO DATA MODEL & ONTOLOGY DEVELOPMENT

EN ISO 12006-2 – Organization of information about construction works - Part 2: Framework for classification

Construction assets are complex objects, e.g. buildings are constructed with walls, which are made up of various products. The most common technique to manage such complexity is classifying items. A classification system enables effective communication on various subjects by presenting sets of concepts that help simplify the complexity of the topic to a manageable level for its users. EN ISO 12006-2 has been developed to (1) establish a framework for the development of built environment classification systems and thus, (2) facilitate the exchange of information between applications throughout the life cycle of construction works, including buildings and civil engineering projects. The standard recommends a systematic approach to structuring terminology for the built environment in an appropriate manner. EN ISO 12006-2 outlines a set of suggested classification tables for various information object classes and is intended for the usage by organisations developing these systems and tables. However, it does not offer a fully operational classification system or the actual contents of the tables, it only provides illustrative examples. EN ISO 12006-2 is referred in EN ISO 19650-1 which states that “the classification of objects must comply with the principles of EN ISO 12006-2”. EN ISO 12006-2 also establishes the foundation for the development of a trustworthy digital content as described in EN ISO 23386 and EN ISO 23387. Actually, the term ‘construction object’ used in EN ISO 23387 is derived from EN ISO 12006-2. Specifically, EN ISO 23387 recommends applying the principles of EN ISO 12006-2 for developing and managing construction objects within data dictionaries.

Please note that, at the time of writing this report, EN ISO 12006-2 is undergoing revision.

EN ISO 12006-3 – Organization of information about construction works - Part 3: Framework for object-oriented information

EN ISO 12006-3 specifies a language-independent information model that can be utilized for the creation of data dictionaries. It allows for the referencing of classification systems, information models, object models, and process models within a unified framework.

Practically, EN ISO 12006-3, also known as IFD (International Framework for Dictionaries), outlines an information model that enables the development of multiple data dictionaries based on the same model. This ensures interconnectivity and interoperability between data dictionaries. To simplify and support the implementation of dictionaries based on this framework, the standard includes a UML model and XML schema as annexes. Additionally, an API specification is added as an annex to standardize and define the minimum functionality to extract and exchange data between dictionaries based on the standard.

EN ISO 23386 – Methodology to describe, author and maintain properties in interconnected data dictionaries

The goal of EN ISO 23386 is to establish a common framework for uniquely describing properties in data dictionaries and managing them, thus, facilitating the high-quality exchange of construction data among various industry stakeholders for diverse applications within different digital tools.

In straightforward terms, the current construction industry business environment necessitates the digital communication of characteristics related to products, systems, components, or processes. In the digital realm, these characteristics are referred to as “properties” of different objects. To guarantee that these properties are machine-interpretable, adaptable, and consistent (enabling their use in various software applications), an agreed-upon method is required for creating these properties. This is precisely the

purpose of EN ISO 23386. Additionally, EN ISO 23386 ensures that this framework establishes the foundation for trustworthy content in data dictionaries through the implementation of a governance process.

Properties and attributes

An object property contains essential metadata known as attributes. These attributes, such as name, definition, translations to different languages, units, values, constraints, serve as descriptors attached to the property. EN ISO 23386 provides a comprehensive list of attributes, ensuring consistency for information exchange and data governance. One of the crucial attributes associated with properties is the Globally Unique Identifier (GUID) which ensures the facilitation of automated processes and data exchange. GUIDs enable machines to identify and differentiate between properties, and this enables the streamlining of data handling, retrieval, and interoperability across digital tools and systems. EN ISO 23386 also introduces a mechanism to group properties, known as 'group of properties', which is also equipped with a list of attributes including a GUID. Groups of properties can serve various purposes, such as grouping properties based on a source document or grouping properties specific to a particular object of interest.

Data dictionaries and data governance

For enabling universal usability across systems, properties and their attributes shall be stored in data dictionaries. Data dictionaries are structured repositories that store information about data elements, their attributes, and relationships. They provide a standardized and centralized way to manage and organize data, ensuring consistency, accuracy, and interoperability across various systems and applications. However, the crucial aspect is not just storage but also ensuring consistent and reliable digital content through governance processes as recommended by the standard. In simpler terms, the standard outlines guidelines for establishing a governance network, defining roles and responsibilities for experts involved in creating properties and attributes.

EN ISO 23387 – Data templates for construction objects used in the life cycle of built assets - concepts and principles

EN ISO 23387's goal is to further enhance the utilization and reusability of data by providing a framework for a data model to group the properties created following EN ISO 23386, and stored and governed in data dictionaries, also known as a data template. Data templates address specific information needs, such as determining the data required for issuing a Declaration of Performance and Conformity (DoPC) in accordance with the Construction Products Regulation (CPR). Manufacturing associations can utilize data templates to establish a common communication method for sharing information about their members' products. Additionally, clients/owners can use them to define their Asset Information Requirements (AIR) as defined in EN ISO 19650 standard series. Practically, data templates offer a mechanism for defining standardized, reusable sets of properties within a specific semantic context. Data templates also have their own attributes, including a description, reference document, GUIDs, and others, also following the framework provided by EN ISO 23386. However, a crucial attribute of a data template is its relationship to another concept in the data dictionary known as a 'construction object'. As per EN ISO 12006-2, a construction object is "any part of a perceivable or conceivable world, in the context of a construction process". The construction object is the concept that specifies the specific item for which the properties in the data template are collected and relevant for, such as a window or mineral wool.

It's important to clarify that EN ISO 23387 exclusively offers the data model and rules. The specific terms, including property names and details about their attributes, originate from product standards such as harmonized technical specifications, European Assessment Documents (EAD), horizontal standards, and other relevant sources. In essence, EN ISO 23387 serves as a framework for "translating" the content of standards into a common digital language, facilitating the creation of trustworthy content for data dictionaries.

After a data template is developed, it is then utilized as a placeholder for actual data, i.e. it gets populated with values. Once this process is completed, the data template transforms into a data sheet, which can be understood as a digital information sheet for the object of interest. It's important to note that EN ISO 23387 does not offer technical guidance on the development and deployment of data sheets.

Please note that, at the time of writing this report, EN ISO 23387 is undergoing revision.

EN ISO 22057 – Data templates for the use of environmental product declarations (EPDs) for construction products in building information modelling (BIM)

The choice of environmentally friendly building materials is a crucial aspect of designing sustainable assets. Thus, having trustworthy and verifiable environmental data on construction products is vital for assessing and overseeing the environmental impact of built assets.

By far, managing environmental data for construction products has been somewhat siloed, necessitating numerous manual data entry points. To address this issue, EN ISO 22057 has adopted data templates, developed, and managed within data dictionaries, to streamline the handling of data from Environmental Product Declarations (EPD).

In contrast to previous digitalization initiatives, this standard aims at providing a data model suitable for both EPD data and generic Life Cycle Assessment (LCA) data. Moreover, the model's scope extends beyond impact indicators, encompassing background information from scenarios. This inclusion holds significant importance for contextualizing EPD data, thus enabling the recalculation of EPD data to align with specific building scenarios, such as the precise distance and method of transportation of the product from the factory to the construction site.

It's important to emphasize that EN ISO 23387 provides the data model and rules for developing a data template. In contrast, EN ISO 22057 leverages these to provide a ready-to-use data template, completed with properties and all relevant attributes. Utilizing this ready-to-use data template, users have the capability to generate an EPD data sheet, which, essentially, is a digital machine – interpretable EPD. This digital EPD can then be integrated into BIM and other digital workflows. As highlighted in the EN ISO 23387 review, data templates are essentially a data model, and similarly, the terms in EN ISO 22057 are derived from Product Category Rules (PCR) standards.

6.2 INDUSTRY DATA MODEL & ONTOLOGY DEVELOPMENT

ILCD+EPD data format

An XML based data and metadata format, co-developed by and endorsed by the main EPD program operators throughout Europe, under the InData initiative. The format builds upon the European Commission's ILCD format made for LCA, while notably leaving out a number of elements, and adding a set of others to be able to transport all relevant EPD information as well. The currently ongoing next development steps aligns with the work of EN ISO 22057 while it otherwise goes beyond.

Define Hub¹⁶

Define is a data dictionary that enables users to author, manage, and share Data Templates. Define Data Dictionary is a solution for the construction industry managed by the industry itself. A management board comprising various construction industry actors is responsible for all decisions concerning the development of the tool. Cobuilder is the operator who integrates the technical requirements and the international standards responsible for working with data.

Define Hub consists of open free of charge data dictionaries published by associations and national working groups. Currently there are European Data Dictionary, UK Data Dictionary, Norwegian Data Dictionary, Swedish Data Dictionary and Portuguese Data Dictionary.

¹⁶ <https://definehub.com/en/>

CEI-Bois initiative TIMBIM (data dictionary)¹⁷

In March 2021, the European Confederation of Woodworking Industries, CEI-Bois, initiated a pilot project together with Cobuilder to help manufacturers digitize their data and make it available in a machine-readable and standardized format.

Below and as part of the pilot project, the CEI-Bois initiative TIMBIM has created a common data dictionary and common data templates that are based on relevant harmonized product and test standards and are applied across all European countries

Since 2023 this content is also published on the international Data dictionary bSDD of buildingSMART international.industry-dictionary for products in wood 1.0.0 - <https://search.bsdd.buildingsmart.org/uri/cei-bois.org/wood/1.0.0>

Based on this template, Lignumdata has created a richly illustrated database of over 4000 wood species and thus offers an insight into the great variety of ligneous plants which is available at <https://lignumdata.ch/?page=holzart&lang=en>

buildingSMART Data Dictionary¹⁸

bSDD is a collection of interconnected data dictionaries with definitions of terms to describe the built environment. The service is provided by buildingSMART for free, to enable easy access from all software solutions. The content is published by independent organizations, spanning international classifications, national standards and company-specific agreements.

ETIM¹⁹

ETIM is an international classification standard for technical products which also includes electronics finished goods. The ETIM product classification model provides the structure for standardized (technical) product data exchange between parties. The standard facilitates the listing and publishing of complete product data with the aim of making it easier to identify the technical products required by the user. ETIM has a flat organizational structure, and the main focus is the definition of product classes and their features which are structured according to their importance as well as key aspects. A feature is clearly described by description, feature type, unit and/or value.

ETIM originated in the electrical and electromechanical sectors, where it has reached a high level of maturity. In comparison, the part for building materials is still developing and is less mature, but it is improving and expanding.

ETIM 9.0 has been also published in the bSDD

¹⁷ <https://www.cei-bois.org/digital-templates>

¹⁸ <https://search.bsdd.buildingsmart.org/>

¹⁹ <https://prod.etim-international.com/class>

7 EXISTING DATA MODELS & ONTOLOGIES: TYRE PRODUCTS

Global Data Service Organisation for Tyres and Automotive Components (GDSO) is an international non-profit association founded in 2022, by leading tyre industry brands. GDSO's purpose is being the responsible body for the development and management of a global tyre data management system, facilitating the secure and standardised access to tyre-related data from digital systems. GDSO's Tyre Information Service provides tyre data defined by the DATA REFERENTIAL attribute set.

8 CONCLUSION

This report has explored the critical role of semantic interoperability in enabling an effective Digital Product Passport (DPP) system. By examining existing data models, ontologies, and classification schemes across key sectors—electronics, textiles, tyres, and construction—we have highlighted both the opportunities and challenges in achieving a harmonized data exchange framework.

A key takeaway is that while semantic anchors and ontologies provide a foundation for interoperability, reliance on a single semantic standard can create conflicts rather than resolving them. To avoid the single semantic anchor trap, the DPP system must be designed with semantic flexibility, ensuring it can integrate with multiple existing ontologies and industry-specific data models.

Furthermore, this report underscores the need for ongoing collaboration and expert input to refine and expand the interoperability framework. As the DPP evolves, continuous engagement with standardization bodies, industry stakeholders, and regulatory authorities will be crucial in ensuring that semantic interoperability remains adaptable and effective across diverse product value chains.

This iteration has focused on mapping the state of the art of existing ontologies and data models across key sectors, identifying gaps and current data models used by pilots. The next phase, Iteration 2, will build upon these findings to develop final versions of sectoral ontologies, ensuring semantic interoperability between the Lighthouse pilots. Moving forward, efforts will be directed toward refining sector-specific ontologies, aligning them with regulatory requirements, and facilitating seamless data integration across the DPP ecosystem.

9 ANNEX A – ONTOLOGY REQUIREMENTS SPECIFICATION FOR THE DPP CORE ONTOLOGY EXTENSION FOR ELECTRONICS

1. Introduction

This specification serves as a guide for the development of the DPP ontology extension for electronics.

We use the term “DPP ontology extension for electronics” when referring to the extension for this product group of the CIRPASS-2 “DPP Core Ontology”.

In terms of specific product groups, the ESPR does not define electronics as a product group, which is a CIRPASS-2 product group naming. The ESPR product groups are IT products and Energy related Products, which includes electronics, which are defined under article 18 of the ESPR as:

- (j) energy related products for which ecodesign requirements are to be set for the first time or for which existing measures adopted pursuant to Directive 2009/125/EC are to be reviewed under this Regulation; and
- (k) information and communication technology products and other electronics.

In the context of this DPP the ontology refers to both IT products, electronics and Energy related Products such as appliances. These categories also cover the pilots in CIRPASS-2 under ‘electronics’ that includes audio equipment, tumble dryers, vacuum cleaners, and IT equipment incl. Laptops.

The purpose of the DPP ontology extension for electronics is to provide shared sector specific vocabulary for DPP data to ensure semantic interoperability for electronics specific product information management IT systems linked to DPPs. Consequently, the DPP ontology extension for electronics is related to electronics specific DPP data.

The extension will integrate with the DPP core ontology whose purpose is to provide shared cross-sectorial vocabulary for sector-specific DPP data to ensure semantic interoperability of DPP system, as identified in Task 4.2.

Similar to the DPP core ontology, the DPP ontology extension for electronics is defined by the CIRPASS-2 consortium as a shared vocabulary that presents machine-interpretable definitions of basic concepts and the relationships between them in the electronics domain.

This document is consumed as an input for the DPP ontology extension for electronics design.

2. Methodology

The approach follows the methodology set in Task 4.2 using the Modular Ontology Modeling (MOMo) methodology, described by Shimizu et al. [Shimizu 2023]. According to the MOMo methodology workflow, ontology requirements specification document delivers the results of the following steps:

1. Identifying ontology use cases, resulting in a set of use case descriptions (brief textual summaries of use scenarios). Use cases are also called use scenarios in the literature as in [Grüniger and Fox 1995].
2. Gathering of competency questions (CQ), resulting in a list of CQs. Competency questions were first introduced by Grüniger and Fox [Grüniger and Fox 1995] for enterprise ontologies modelling.
3. Identifying key notions. The result is a list of core terms.

In addition, principles of ontology engineering from [Gómez-Pérez et al 2004] are used to provide scope and goal of the ontology, a list of non-functional requirements and ontology representation languages.

The DPP ontology extension for electronics will be developed following the iterations of the CIRPASS-2 DPP core ontology, so as to interlink both.

3. Domain and Scope

The purpose of the DPP ontology extension for electronics is to provide shared sector specific vocabulary for DPP data, covering IT products and Energy related Products, as per relevant product groups under ESPR.

Since the EU delegated act for these product groups will be provided at a significantly later date than the ontology development works, it will not be possible to consult or consider this delegated act for the works to be carried out.

The general needs for a shared sector specific vocabulary, and linked semantic interoperability, will be to enable interoperability for electronics specific product information management IT systems initiatives for linkage to DPPs, which covers both IT products and Energy related Products. Consequently, the DPP ontology extension for electronics is related to electronics specific DPP data.

Specific product information management IT systems initiatives relevant for the DPP ontology extension for electronics include:

- **EPREL database**, the European Product Registry for Energy Labelling that covers product characteristics and energy labels for 25 Energy related Products <https://eprel.ec.europa.eu/screen/home.1>
- **The PI Standard**, a European wide standard (and data structure) developed by APPLIA and several of its members, that covers a concise listing of features for a wide range of small and large electrical appliances, for brand/retail purposes <https://www.picertified.com/>.
- **Global Electronics Council EPEAT registry**, a global registration and certification offering for IT products covering environmental and circularity product characteristics <https://www.epeat.net/#search>.
- **E-Class classification** reference standard, used for classification of products and service, a standardised product naming reference that includes Energy related and IT products.
- **IEC 62474 Declarable Substance Database**, used globally for declaring substances in EEE as required in legislations in different geographies. <https://rohs.ca/iec62474/>

4. Ontology Use Case Descriptions

In ontology specification methodologies, including MOMo [Shimizu 2023], use case descriptions are brief textual summaries of ontology use scenarios that help to understand the problem to be addressed by the ontology, including potential future extensions.

The following ontology use case descriptions are identified for the ontology extension for electronics (sector-specific ontology):

- The sector-specific ontology can be used to describe in DPP applications, and automate the process of creating from DPP data structures, EEE specific product data descriptions as defined in the EPREL database for interoperability.
- The sector-specific ontology can be used to describe in DPP applications, and automate the process of creating from DPP data structures, EEE specific product data descriptions as defined in the PI standard for interoperability.
- The sector-specific ontology can be used to describe in DPP applications, and automate the process of creating from DPP data structures, EEE specific product data descriptions as defined in the EPEAT database for interoperability.

- The sector-specific ontology can be used to describe in DPP applications, and automate the process of creating from DPP data structures, EEE specific product data descriptions as defined in the E-Class classification standard for interoperability.
- The sector-specific ontology can be used to describe in DPP applications, and automate the process of creating from DPP data structures, EEE specific product data descriptions as defined in the IEC62474s material & substance declaration standard for electrical and electronic products for interoperability.
- The sector-specific ontology will be linked to the CIRPASS-2 DPP core ontology and will inherit it. Thereby, the use cases as described in Task 4.2 can also be utilised.
- The established DPP Core Ontology can be used by application developers responsible for building or integrating DPP applications. Using a shared vocabulary helps to build interoperable, and regulation-compliant solutions.

5. Ontology Representation Languages

The approach is inherited from Task 4.2 to align fully with the methodology and make the sector-specific ontology simple to extend relative to the CIRPASS-2 core ontology.

The CIRPASS-2 core ontology language representation is as follows:

- On the conceptual level, the DPP core ontology is represented using a graphical modelling language like the Unified Modelling Language (UML).
- The DPP core ontology will be created in Web Ontology Language (OWL). OWL is W3C standard that enables rich ontology representation and ontology consistency checks using Description Logic (DL) reasoners (e.g., OWL-DL reasoner Pellet). In addition, using OWL for ontology representation makes it possible to use the semantic query language for RDF, SPARQL, as well as Shapes Constraint Language (SHACL) that is a language for validating RDF graphs against a set of constraints. Both, SPARQL and SHACL, are W3C standards.
- If necessary, OWL ontologies can be converted to RDF, Turtle formats (W3C standards) as well as to JSON-LD (<https://json-ld.org/>) that is a Linked Data format based on JSON.

6. Ontology Requirements

Non-functional requirements

Non-functional requirements follow from the Task 4.2 core ontology non-functional requirements, for which the associated deliverable can be consulted.

Functional Requirements: Competency Questions

The functional requirements (behaviour of the ontology), following the MOMo methodology, are described through Competency Questions (CQ).

Competency questions are examples of natural language queries for which the DPP ontology extension for electronics should provide answers after the ontology is populated with data. The list is not exhaustive.

Note that the first three of these CQs are sector-specific duplicates of the CQ in Task 4.2 for the core ontology, where they relate to sector specific information. Since the core ontology is still under development it is not clear at present if the inheritance of it will be sufficient to cover all requirements for information management specific to IT products and/or energy related products, hence these are included here also.

CQ1: What are the values of all or selected mandatory attributes of a DPP required by ESPR delegated act for IT products and/or Energy Related Products at the granularity level specified in the delegated act (model, batch or item level) or at the model level?

CQ2: What are the values of all or selected performance parameters of the IT and/or Energy Related product? (ESPR Art 7 (2b), (i))

CQ3: What type of digital instructions are available for the IT or Energy Related product? (ESPR Art 7 (2b), (ii) and (iii), Art 7 (5d) and Art 7 (5e)).

CQ4: What are the values contained in the physical energy label of the product contained in the EPREL database?

CQ5: What are the values for the general information parameters contained in the EPREL database product data entry?

CQ6: What are the values of all product parameter contained in the EPREL product information sheet for the product?

CQ7: What is the classification of the product based on the E-CLASS classification as setup in the Digital Product Passport?

CQ8: What are the values of all product parameters contained in the PI-standard information sheet for the product?

CQ9: What are the values of all product parameter contained in the EPEAT registry information sheet for the product?

CQ10: What are the EEE specific material, substance and elemental composition values of the product as defined based on the IEC62474 standard?

7. Definitions of Core Terms

Definitions of horizontal terms are covered under Deliverable 4.2 of CIRPASS. The below are specific terms related to electrical and electronic products.

Energy Related Products. An energy related product (ErP) is a product that uses energy or has an indirect effect on energy consumption

IT products. IT products are computer hardware and software products that are used to process information.

Electrical and Electronic Equipment (EEE). Equipment that uses electric currents or electromagnetic fields to work.

8. Ontology Requirements Gaps

The main requirement gap relates to the product types which will fall under the ESPR delegated acts for IT products and Energy related Products. For example, is lighting or are electric motors covered, since these are also covered under the EU ErP directive? Once the full scope of product coverage becomes clearer based on the first working plan to be released in the 1st half of 2025 by the EU commission, this scope gap can be better addressed.

9. References

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